

Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies
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PROFESSIONAL PROFILE

M.Sc. Quantum Technologies student at TU Braunschweig with a B.Sc. in Computer Engineering (AI specialization). Applied projects in AI-assisted Python/ML, computer vision and sensor analysis; Master's work connects quantum computing, coding, photonics and devices.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering
Oct 2024 – Present
2 yrs 1 mo
Next: Internship → potential Thesis

Technische Universität Braunschweig — Braunschweig, Germany | 70/120 ECTS completed
Focus: Quantum Computing • Information & Coding Theory • Photonics & Devices

B.Sc. Computer Engineering (AI specialization)
Sep 2019 – Feb 2024
4 yrs 6 mos

Azad University of Mashhad — Mashhad, Iran
Focus: Software & AI • Computer Vision • Embedded Systems & Databases

EXPERIENCE

Independent Software & AI Projects — Münster, Germany
Oct 2025 – Present
1 yr

Software & AI Project Developer (AI-Assisted)
- Defined and evaluated AI-assisted Python/ML, computer-vision and sensor-data projects; integrated components and checked outputs. Co-developing Career OS workflows. **Next focus:** Quantum computing and ML applications.

Iran Khodro (IKCO) — Iran's largest automaker — Mashhad, Iran
Apr 2023 – Sep 2023
6 mos
Top 1

Computer Vision Developer — Bachelor's Thesis (Pouyesh Program)
- **Selection:** Ranked 1st of 50 top engineering students from Azad and Ferdowsi Universities.
- **Industry challenge:** Developed a MATLAB vehicle detector for an assigned traffic-video problem.
- **Thesis & outcome:** Project became my Bachelor's thesis; invited to join IKCO R&D.

Ferdowsi University of Mashhad — Mashhad, Iran
Mar 2023 – Aug 2023
6 mos

Project Contributor - Anodyne (University-Based Startup)
- **Project support:** Supported Anodyne dental-imaging presentations, stakeholder outreach and medical/dental industry events.

Azad University of Mashhad — Mashhad, Iran

Teaching Assistant - Theory of Formal Languages and Automata
Sep 2023 – Jan 2024
5 mos
- **Teaching & assessment:** Led weekly exercise/problem-solving sessions for approximately 20 students and evaluated assignments and exams under course supervision.

IT Support
Oct 2022 – Aug 2023
11 mos · Volunteer
Jul – Sep 2022
3 mos
- **Hardware, networking & cameras:** Troubleshot and upgraded PCs, configured Windows/LAN/Wi-Fi, supported rack cabling and connected camera systems across university departments.

SELECTED MASTER'S TECHNICAL WORK | Portfolio

COMMUNICATION, INFORMATION & CODING THEORY | Details
2025–2026

Scientific Areas: Information & Coding Theory; Error-Correcting Codes; Network Information Theory; Quantum Communication. **Selected Work:** Participated in a collaborative MATLAB coding-theory lab on Hamming/cyclic codes and errors versus SNR; submitted information-theory exercises and prepared presentations on state-dependent coding, Bloch-state characterization and CHSH communication.

COMPUTING, ALGORITHMS & OPTIMIZATION | Details
2025–2026

Scientific Areas: Quantum Information & Algorithms; Quantum Circuits; Quantum Programming (Qiskit); Introductory Quantum Machine Learning. **Selected Work:** Completed assessed theoretical coursework on quantum states, measurement, entanglement and QFT; organized AI-assisted Qiskit learning examples on GHZ, Bernstein–Vazirani and introductory VQE.

QUANTUM DEVICES & PHYSICAL SYSTEMS | Details
2025–2026

Scientific Areas: Nonlinear Photonics; Semiconductor & GaN Device Physics; Quantum Hardware Foundations; Nano-Optics & Plasmonics; Electromagnetic Fields. **Selected Work:** Prepared twelve nonlinear-photonics analytical submissions and technical presentations on crystalline dislocations and published Sn-nanobar plasmonic-metasurface/FDTD findings (literature review).

Updated: 17 October 2026

TECHNICAL SKILLS

Programming	Python; C; C++; MATLAB
Machine Learning & Computer Vision	PyTorch; scikit-learn
Scientific Computing & Signal Processing	NumPy; SciPy; pandas; Matplotlib
Embedded Systems	ATmega32
Development Tools & Databases	Git; GitHub; GitLab; SQL

SELECTED TECHNICAL PROJECTS

AI SYSTEMS & WORKFLOW AUTOMATION — 2026

Career OS - AI-Assisted Academic & Career Workflow Platform (Link)

Google Apps Script; Google Drive API; Gemini/Groq; GitHub Actions

- Co-designed a privacy-first academic/career workflow platform, including the personal portfolio website as its managed public-profile subsystem; defined privacy/evidence rules and guided AI-assisted work on AI context, Drive ingestion, queue/retry automation, API integrations and CI checks.

MACHINE LEARNING & COMPUTER VISION — 2025

MRI Segmentation Inference Dashboard (Link)

PyTorch; Streamlit

- Integrated pretrained segmentation inference with preprocessing, weight loading, masks, overlays, and area visualization.

Gas Classification Pipeline and Simulation Dashboard (Link) — 2025

Python; scikit-learn; RandomForest; Streamlit

- Built statistical sensor-window features and a six-gas classification dashboard with class-probability output.

Activity Recognition ML Comparison (Link) — 2025

Python; PyTorch; scikit-learn; Poetry

- Compared RandomForest and 1D-CNN for sensor activity recognition in a Python project including lint/test CI checks.

Vehicle Detection & Traffic Video Analysis (Link) — 2023

MATLAB; Computer Vision; Image Processing

- Built separate MATLAB detection paths: HSV-based scene classification, Gaussian-mixture foreground segmentation and morphological filtering for daytime; binary thresholding and connected-component analysis for nighttime vehicle counting.

SIGNAL PROCESSING — 2023

Stereo Sound Direction (Link)

MATLAB; Signal Processing

- Analyzed stereo-audio window energy to identify the dominant left/right direction.

EMBEDDED SYSTEMS — 2021–2022

ATmega32 Digital Clock (Link) | ATmega32 Digital Safe (Link)

Embedded C; ATmega32; 7-segment display; matrix keypad; LCD

- Implemented coursework projects for a multiplexed digital clock and a keypad/password digital safe.

HONORS

B.Sc. Computer Engineering: Ranked 5th of 131 students	**2024 · Top 5**
Iranian University Entrance Exam (Konkoor, Mathematics): ~10,000 of ~144,500	**2018 · Top 7%**
Khorasan Razavi selective-school Entrance Exam: 12 of ~27,000	**2014 · Top 1%**

LANGUAGES & CERTIFICATES

Languages

German - A2 course completed; B1.1 in progress
English - TOEFL iBT 95/120
Persian - Native

Training (non-degree)

Sharif University of Technology (PsiKet Academy) - Quantum Programming Languages (ongoing)	2026
AriaQuanta Institute - Quantum Machine Learning (Intro)	2025
Kaggle - Machine Learning; Introduction to Programming Using Python	2025
Azad University, Mashhad - Data & ML Bootcamp - 96 hours	2023
Jahad Daneshgahi, Khorasan Razavi - Python Programming - 40 hours	2021